HOW TO FIND INTEGRATION BY PARTIAL FRACTION

IB MATH PUNE

$$\int \frac{3x+5}{x^2-1}\,dx$$

$$\frac{2}{x-1}+\frac{1}{x+1}$$

5 STEPS

1. IDENTIFY & FACTORISE
2. SET UP PARTIAL FRACTIONS
3. FIND THE CONSTANTS
4. INTEGRATE EACH TERM
5. WRITE THE FINAL ANSWER

WHATSAPP
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YOUTUBE CHANNEL
https://www.youtube.com/@IBMATH-in

AHL 5.15

Content	Guidance, clarification and syllabus links
Derivatives of $\tan x$, $\sec x$, $\operatorname{cosec} x$, $\cot x$, a^x, $\log_a x$, $\arcsin x$, $\arccos x$, $\arctan x$.	
Indefinite integrals of the derivatives of any of the above functions. The composites of any of these with a linear function.	Indefinite integral interpreted as a family of curves. **Examples:** $\int \frac{1}{x^2+2x+5}\mathrm{d}x = \frac{1}{2}\arctan\frac{(x+1)}{2} + C$ $\int \sec^2(2x+5)\mathrm{d}x = \frac{1}{2}\tan(2x+5) + C$
Use of partial fractions to rearrange the integrand.	$\int \frac{1}{x^2+3x+2}\mathrm{d}x = \ln\left\lvert\frac{x+1}{x+2}\right\rvert + C$ **Link to:** partial fractions (AHL1.11)

Question (1): Integrate with respect to 'x':

$$\frac{2-x}{2x^2-5x-3}$$

Let $$\frac{2-x}{2x^2-5x-3}=\frac{2-x}{(x-3)(2x+1)}=\frac{A}{x-3}+\frac{B}{2x+1}$$

$$2x^2-5x-3$$
$$=2x^2-6x+x-3$$
$$=2x(x-3)+1(x-3)$$
$$=(x-3)(2x+1)$$

$$\therefore 2-x=A(2x+1)+B(x-3) \quad \text{—— (i)}$$

Put x = 3 in (i), We get

$$2-(3)=A(2(3)+1)+B(0)$$

$$\therefore A=-\frac{1}{7}$$

Put x = – 1/2 in (i), We get

$$2-\left(-\frac{1}{2}\right)=A(0)+B\left(-\frac{1}{2}-3\right) \Rightarrow B=-\frac{5}{7}$$

$$\therefore \frac{2-x}{2x^2-5x-3}=\frac{-1/7}{x-3}+\frac{-5/7}{2x+1}$$

$$\therefore \frac{2-x}{2x^2-5x-3} = \frac{-1/7}{x-3} + \frac{-5/7}{2x+1}$$

$$\therefore \int \frac{2-x}{2x^2-5x-3}dx = \int \frac{-1/7}{x-3}dx + \int \frac{-5/7}{2x+1}dx$$

$$= -1/7 \ln|x-3| - 5/7 \frac{\ln|2x+1|}{2} + C$$

$$= -\frac{1}{7}\ln|x-3| - \frac{5}{14}\ln|2x+1| + C$$

Question (2): Integrate with respect to 'x':

a Write $\dfrac{6x^2+x-19}{(x+3)(x-1)^2}$ as the sum of partial fractions in the form $\dfrac{A}{x+3}+\dfrac{B}{x-1}+\dfrac{C}{(x-1)^2}$.

b Hence find $\displaystyle\int \frac{6x^2+x-19}{(x+3)(x-1)^2}\,dx$.

(a) Let $\dfrac{6x^2+x-19}{(x+3)(x-1)^2}=\dfrac{A}{x+3}+\dfrac{B}{x-1}+\dfrac{C}{(x-1)^2}$.

$$\therefore 6x^2+x-19=A(x-1)^2+B(x+3)(x-1)+C(x+3) \longrightarrow (i)$$

Put x = – 3 in (i), We get

$$6(-3)^2+(-3)-19=A(-3-1)^2+B(0)+C(0)$$

$$32=16A \Rightarrow A=2$$

Put x = 1 in (i), We get

$$6(1)^2+(1)-19=A(0)+B(0)+C(1+3)$$

$$\therefore c=-3$$

$$\therefore 6x^2 + x - 19 = A(x-1)^2 + B(x+3)(x-1) + C(x+3) \text{ —— (i)}$$

Put x = 0 in (i), We get

$$6(0)^2 + (0) - 19 = A(0-1)^2 + B(0+3)(0-1) + C(0+3)$$

$$-19 = (2) - 3B + (-3)(3)$$

$$\frac{6x^2+x-19}{(x+3)(x-1)^2} = \frac{A}{x+3} + \frac{B}{x-1} + \frac{C}{(x-1)^2}.$$

$$\therefore B = 4$$

$$\therefore \frac{6x^2+x-19}{(x+3)(x-1)^2} = \frac{2}{x+3} + \frac{4}{x-1} - \frac{3}{(x-1)^2}$$

(b) $$\therefore \int \frac{6x^2+x-19}{(x+3)(x-1)^2}\,dx = \int \frac{2}{x+3}dx + \int \frac{4}{x-1}dx - \int \frac{3}{(x-1)^2}dx$$

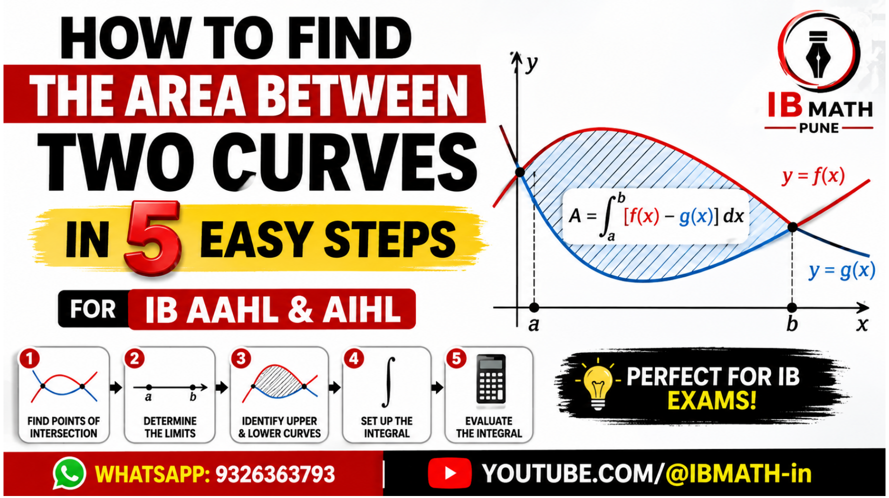
$$= \int \frac{2}{x+3}dx + \int \frac{4}{x-1}dx - \int 3(x-1)^{-2}\,dx$$

must be linear

$$= 2\ln|x+3| + 4\ln|x-1| - 3\frac{(x-1)^{-1}}{-1} + C$$

$$= 2\ln|x+3| + 4\ln|x-1| + \frac{3}{x-1} + C$$

Practice questions PDF link is given in the description of this video.



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